CDSS 198: Data Discovery[1] | Fall 2025

Meetings: Mondays 3-5 pm
GPB 100: Everyone should meet in GPB to begin class meetings
VLSB 2060: Those enrolled in section 005 will relocate to VLSB for team meetings

Instructors: Prof Deb Nolan and Associate Dean Narges Norouzi
Facilitated by: Data Discovery Associate Director Emily Remirez, PhD
Data Discovery Director George Obaido, PhD (Data Discovery Director).

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Course description and goals

The CDSS Data Discovery Program offers Berkeley students an opportunity for research and exploration, using data to understand real-world problems. Teams of students work together

[1] Subject to minor revision. Last updated 10/06/2025

with a mentor in academia, industry, government, or non-profit organizations towards a data-centric goal. Through your project work, you'll be exposed to the challenges of applying your knowledge to messy, real-world problems outside of a curated course environment. You'll see the direct impact of your contribution on your mentor's broader goals. Hone relevant skills, build your resume, and get firsthand exposure to careers working with data.

CDSS 198 is designed to support your work on the project. The hands-on experience with your mentor will develop your practical expertise in the topic; the course will lay the foundation for this work across four overarching, interrelated themes:

1. **Contextualizing your work** within the larger framework of the project and integrating your experience with your studies and professional goals.
2. **Communicating** clearly and genre-appropriately about your data, methods, and interim results with different audiences.
3. **Working as part of a team** towards collaborative learning, project management, and problem-solving.
4. **Developing personal processes** around time management, resourcefulness, and independent work.

Expectations for the project

The details of working together will be specific to your team and should be negotiated and revisited collaboratively. There are a few things we expect of everyone:

- **Weekly project hours:** You should negotiate with your mentor to commit to either 5-7 (3 units) or 7-10 hours (4 units) per week and enroll in the appropriate number of units. This is an average across weeks, not a strict minimum per week. Your mentor shouldn't require you to work on non-instructional days or weekends. If you need to reduce your hours for a given week, you're responsible for communicating with your mentor and teammates with as much advance notice as possible. You may be asked to make up the hours another week.
- **Proactive communication with mentor and team:** Keep your group updated as issues arise (e.g., if you aren't able to complete a task as expected or have a schedule conflict).
- **Academic and intellectual honesty**: Any work you submit to your mentor should reflect your genuine contributions, and you should never take credit for work that isn't

yours. Do not use generative AI for any part of your project without clearing it explicitly with your mentor.

Computing resources

Computing resources are available for you to complete your work on the project. More details can be found at these links:

- DataHub: https://berkeley-cdss.github.io/discovery-handbook/computing/datahub/
- Savio: https://berkeley-cdss.github.io/discovery-handbook/computing/savio/

Please reach out to George if you have questions or concerns about accessing these resources.

Office hours

Emily, George, and Claire are available to meet with you about your project and the course. Office hours are on Zoom by default and can be scheduled in 15-minute increments.

Emily is available to talk about things like CDSS 198 assignments and grading; skills related to social science, humanities, and qualitative research; interpersonal issues; writing and communication; and getting connected to university resources.

George and Claire can help with technical issues. George is excited to talk to you about Data Science, LLMs, choice of ML algorithms/tools, fairness/AI bias, responsible AI, and Data Discovery as a program.

Table 1: Staff office hours

Staff	**Schedule**	**Set an appointment**
Emily	Thursdays 2:30-4, Zoom	Emily's office hours
George	Tuesdays 2-4, Zoom	George's office hours
Claire	Wednesdays 10-12, Zoom	Claire's office hours

Assignments and grading

Your final grade will be calculated based on your attendance record, lab notebook, final presentation (poster), and mentor feedback.

Table 2: Grading breakdown and point distribution

Expectation	**Description**	**% of grade**
CDSS 198 attendance and participation	Attend 75% of class meetings for full credit. Attendance includes engaged participation in group discussions.	45
Lab notebook assignments	Maintain a notebook documenting your progress and submit required near-weekly entries to Gradescope.	20
Final presentation	Present a poster at the Symposium or an approved alternative with both audio and visual components. No standalone written reports will be accepted.	20
Mentor evaluation	Your project mentor will be asked for feedback on your engagement. You will be penalized if they or your teammates report serious concerns about your commitment.	10
Program assessments and surveys	Throughout the semester, you'll be asked to respond to surveys to help us assess Data Discovery as a program. These surveys are confidential and graded on completion.	5

You will be assigned a numeric grade. For Fall 2025, grades will be converted to P/NP (undergrads) or S/US (grad students). The minimum passing grade is set by the university at C- for undergrads and B- for grad students. For Data Discovery, this is defined at 70% and 80%, respectively. A non-passing grade may also be assigned if there are credible reports of particularly egregious lack of participation in project assignments. Students who receive a non-passing grade may be barred from participating in Data Discovery in the future.

Written assignments will be submitted via Gradescope. The course number is **J6E7YX**. Many assignments will be group submissions you complete with your team.

Participation and attendance

Participating in class means showing up every week ready to work with your team on project tasks and discuss the day's topic. It also means paying respectful attention and not being disruptive during lectures or presentations. Class meetings should be thought of as a lab: The work happens here and now, so attendance is crucial.

Attendance will be checked each meeting with a 1-3 question quiz, graded on completion. The quiz will open with a unique link at a different time each meeting. Do not forward the link to people who are not in class; doing so is considered academic dishonesty.

Plan to bring a device (e.g., your laptop) to class every week to respond to the attendance question and work on lab notebook assignments and your project.

Do not come to class if you are ill. You need rest to recover, and you protect the entire community by self-isolating when you might be contagious. You're encouraged to wear a mask to class, especially if you may have been exposed to Covid or another illness.

Team meetings

The second part of each class will be dedicated to team meetings. During this time, you should actively engage with your teammates to make progress on your project. This could include working on project assignments, individually or collaboratively; seeking advice from course staff or other classmates; or working on the week's lab notebook assignment (more below). If you need further technical support, schedule an office hours appointment with George or Claire.

In addition to these in-class meetings, you should be meeting as a team with your project mentor at least once a week. You may also want to schedule a student-only meeting outside of class, depending on your project needs.

Lab notebook

Throughout the semester, you'll maintain a lab notebook documenting your progress. This includes individual or group responses to the week's topic. Assignments are due the Monday

following the relevant class period. Most of these assignments are meant to be brief records or reflections of your team discussion. We encourage you to finish the entry during the class meeting if possible and not invest a lot of time into them outside of class (20 minutes-1 hour).

Submit assignments as PDFs on Gradescope (course code J6E7YX). Each submission should include only the entry for that assignment.

Contribution statements

Every assignment should include a contribution statement. All items must be included, whether or not they apply. For example, if you don't use generative AI, include the statement "No generative AI tools were used for this assignment."

1. Each teammate's contribution. Not everyone will contribute equally to every single assignment, and that's okay! However, it should average out over the semester
2. Any outside sources that were consulted, including other people you talked to
3. A disclosure about whether and how generative AI was used

Lab notebook deadlines and grading

Each of the 10 lab notebook assignments is worth 2 points, for a total of 20. Assignments are based largely on completion. A mostly complete assignment that demonstrates an effort to grapple with the topic will receive full credit. An assignment judged to be incomplete or have some problems will receive half credit, and work that is missing or has serious problems will receive no credit.

Assignments are due on Mondays at 11 pm. There is a 24-hour grace period to turn in work without a penalty. After 24 hours, you will be deducted 1 point. After 48 hours, late work will not be accepted for credit.

Table 3: Late penalties for written assignments

Day submitted (11 pm)	**Max score**
Monday	2 points
Tuesday	2 points
Wednesday	1 point

Thursday or later	0 points

Lab notebook contents overview

Table 4: Summary of notebook assignments

Week	**Assignment**	**Due (11pm)**	**Submission type**
1	1. Mentoring agreement	~~09/15~~ 09/22	Group
2	2. Team collaborative plan	09/22	Group
3-4	3. Describing your project in context	10/06	Group
5	4. Data description	10/13	Group
6	~~5. Problem-solving strategies and resources~~ 5. Project description revision	~~10/20~~ 10/22	Group
7	~~6. Team debrief + mentor feedback reflection~~ 6. Data description revision	10/27	~~Individual~~ Group
8	7. Dear Data self study	11/03	Individual
9	8. Communicating with a non-expert	11/10	Individual
9-10	9. Candidate tables and figures for presentations	11/17	Group
11	~~10. Abstract draft 1~~	11/24	Group

Final presentation

You are required to deliver a presentation about your work this semester with both visual and oral components. The presentation must contain only the original work of your current student team. Your mentor should serve in an advisory capacity, offering support and helping to ensure accuracy. We consider all projects to be “works in progress,” and your project does not need to reach any particular point for you to share it! For example, you could describe a problem you encountered and how you tried to address it.

There are 3 interrelated objectives that every successful final presentation must achieve:

1. **Clarity of message.** You've thoughtfully chosen a topic related to your Data Discovery project and appropriate to present to an audience not directly familiar with your work. You've followed best practices to communicate clearly.
2. **Structure & organization.** Your presentation has an internal structure with a beginning, middle, and end. It's easy to follow, well-balanced, and organized to enhance understanding of your message.
3. **Visual communication.** Visuals (graphs, charts, figures) should feature prominently in your presentation. They're thoughtfully chosen and designed to work together with written and oral communication.

Data Discovery Symposium

All Data Discovery participants are welcome and strongly encouraged to present a poster at the Data Discovery Symposium during RRR week! This is both the most straightforward and most fun way to fulfill the final presentation requirement.

To present at the Symposium, you must submit a registration form (link to follow) by Wednesday, December 3 (link to follow), including an abstract that summarizes your presentation.

Table 5: Fall 2025 Data Discovery Symposium Details	
Date	Wednesday, December 10, 2025
Location	Hearst Memorial Mining Building (HMMB) Lobby
Event time	1:30-3 PM
	2 PM: Keynote (Assoc. Dean Narges Norouzi) 2:15 PM: Presentation of student awards
Presenter time commitment	12:30-4 PM

The Symposium is our chance to connect as a community and celebrate all the work you've done. The event is open to the public. Please invite your project mentor to join us! We also

encourage you to invite your friends, family, classmates, instructors, and anyone else who might like to see your presentation.

Poster printing

Printed posters should be 24 inches x 36 inches (landscape or portrait).

You have the option to print your poster through our group order with a local printer. For the group order, your poster is due *to the printer* **Monday, December 1, 11:59 PM**.

Group order link: https://www.posterpresentations.com/groups/CDSS/cdss-discovery-poster-printing.html

There is a small discount for joining the bulk order, bringing the cost to just under **$34.** Your mentor is expected to pay for your poster printing. You're also welcome to use any other service you prefer. You may be eligible for low cost or free printing from the Jacobs Institute: https://jacobsinstitute.berkeley.edu/jps/

If you order with the group order, Emily will pick up your poster and bring it to the Symposium. If you order your poster from another venue, plan to bring it with you.

Other final presentations

Two kinds of alternate final presentations will be accepted with prior approval. Details about how to seek approval will be provided later in the semester.

1. If you're unable to present a poster publicly during the symposium (e.g., due to confidentiality concerns or a major conflict), you can record a video that includes *both* audio narration and visuals (e.g., a slideshow or digital poster). Your request should include your plan for the presentation and your rationale.
2. If you're giving a comparable presentation elsewhere (e.g., at a conference or to your mentor's organization), you can petition for this presentation to replace your poster. This should include the date, audience, format, and an overview of the content. You're welcome and encouraged to contribute to writing the petition, but we must receive it directly from your mentor for it to be considered.

Mentor Evaluation

Your project mentor will be asked for feedback on your engagement, to be shared both with Data Discovery and with you directly.

You will be penalized if your mentor or teammates report serious concerns about your commitment or engagement. You will not be penalized for things like failure to achieve a desired result in your analysis or deviating from the initial timeline as the project develops.

Program assessments and surveys

You will be asked to complete 5 confidential surveys and self-assessments throughout the semester. Surveys are graded on completion, and your specific answers will not affect your grade. Your honest responses are crucial for evaluating the impact of and improving Data Discovery.

The first survey will set a baseline for where you are now with your current skills and confidence with course topics. This will be used by us and by you to measure your progress over the semester.

Academic honesty and generative AI (genAI)

- As with all assignments, you must abide by the UC Berkeley Code of Student Conduct and Honor Code, which prohibit cheating, plagiarism, and presenting false information.
- We expect that most of your assignments will be a collaborative effort among your team. Include a statement describing each person's contribution to the work.
- If you reference outside sources, acknowledge the source of the information with a citation following any major style guide. Include a link to any source accessed online.
- If you choose to use generative AI (genAI; e.g., large language models, including chatbots like Chat-GPT, Gemini, Claude, etc), **you must:**
    - Include a description of the tool you used and how you used it.
    - Ensure that all ideas come from you and your prompt, with the AI serving as a tool for organizing your ideas and refining your writing.
    - Refer to campus guidance on using genAI in your work.
- Do not misrepresent any work and ideas from other sources as your own. Doing so constitutes plagiarism. This includes your Data Discovery teammates, your GSI, Wikipedia, a journal article, a podcast, and genAI.

- Do not present false information. You are solely responsible for any inaccuracies, including those copied from genAI.
- Do not use genAI in your project work without explicit permission from your project mentor.

Schedule

Table 6: Week-by-week schedule for the semester

Week	Date	Topics	Assignment (due the following week)
		Unit 1: Launching your team collaboration	
-2	08/18		
-1	08/25		
0	09/01	**Team formation - *no class***	
1	09/08	**Program orientation** ● Understanding program and course expectations ● Setting expectations and procedures for your Data Discovery project ● Getting to know your team	Mentoring agreement
2	09/15	**Working in teams** ● Best practices for teamwork ● Setting expectations and norms for your collaboration	Team collaborative plan
		Unit 2: Data research basics	
3	09/22	**Data life cycle** ● Review the stages of a data research project ● Understand where in the DS life cycle your project work fits in	*Begin work on context description. Do not submit yet!*
4	09/29	**Project life cycles** ● Panel presentation on project management	Describing your project in context

		• Understand how your contribution to the project fits into the overall	
5	10/06	**Data descriptions** • Features and characteristics of data • Thinking about the particular data you're working with	Data description
		Unit 3: Identifying and addressing challenges with data	
6	10/13	**Resourcefulness and problem-solving** • Panel presentation on strategies for responding to challenges • Identifying strategies and resources to use in addressing problems • Developing resilience and perseverance	Revising project descriptions
7	10/20	**Evaluating progress and regrouping** • Debriefing with your team • Documenting and reflecting on feedback from your mentor	Revising data descriptions
8	10/27	**Time management and personal habits** • Evaluating your current workflows • Learning strategies and resources available across campus	Self study + Dear Data visualization (*individual assignment*)
		Unit 4: Communicating about your work	
9	11/03	**Reporting on projects** • Abstracts, conclusions, and citations	Communicating with a non-expert (*individual assignment*)
10	11/10	**Communicating with visualizations** • Panel presentation on creating and describing data visualizations • Planning visualizations for your presentation	Preparing figures and tables for presentation
11	11/17	**Designing posters** • Understanding the goals and basics of a scientific poster • Deciding what to include in your poster	

		Unit 5: Synthesizing your work into a final presentation	
12	11/24	**Symposium info session - *virtual meeting*** ● Expectations for the Symposium ● *Meeting will be recorded, and attendance is optional*	
13	12/01	**Verbal communication** ● Talking about your Data Discovery experience for future opportunities ● Practicing oral presentations for the Symposium *Project work ends, last class meeting*	Deadline to submit poster to group order 11:59pm
14	12/08	**Symposium on December 10**	
15	12/15		

Accommodations and resources

Our well-being is a prerequisite to our learning. If you're struggling this semester academically or personally, we can work together to mitigate the effects. As instructors, there are some forms of violence and abuse that we are legally required to report. Whenever we are not required by contract to report information, we will always respect your privacy.

Disability accommodations

Diverse perspectives improve the learning environment for everyone, and all students have a right to participate equitably in class. Students with disabilities are legally entitled to reasonable accommodations to make this course and program accessible. The Disabled Students Program (DSP; https://dsp.berkeley.edu/home) is the campus office responsible for authorizing disability-related academic accommodations.

An *accommodation* mitigates access barriers without interfering with the core functions of learning or assessment. Accommodations are a collaborative effort that involves students, instructors, and disability specialists. Examples of accommodations include, but are not

limited to, extra time or extensions for assignments; a quiet or separate environment for testing; excused absences; adaptive furniture; captioning, transcription, or note-taking services; alternate formats for course materials; and accompaniment by a service animal.

If you anticipate needing disability accommodations this semester, please work with your DSP specialist to send your letter to us as soon as possible. You are responsible for communicating with us about using your accommodations. For example, if you have an accommodation for extensions, you must request the extension and set a new due date in advance.

Confidential resources

- For SVSH and intimate partner violence concerns, contact **PATH to Care:** care.berkeley.edu; for urgent support, call 510-643-2005
- For mental health concerns, The **Tang Center** (https://uhs.berkeley.edu/caps) provides both individual and group counseling for students, and has drop-in hours during the week (https://uhs.berkeley.edu/counseling/urgent)
- For other concerns, such as interpersonal conflicts with someone else on campus, call the **ombuds office** (https://sa.berkeley.edu/ombuds)

Other campus resources

- **Accessing resources:**
  - Basic Needs Center: https://basicneeds.berkeley.edu/pantry
  - Open computing facility: https://www.ocf.berkeley.edu/
  - Student Technology Equity Program: https://studenttech.berkeley.edu/step
  - Campus-provided software downloads: https://software.berkeley.edu/
- **Community & campus spaces**
  - Centers for Educational Justice and Community Support: https://cejce.berkeley.edu/
    Links to Undocumented Students' Program, Gender Equity Resource Center, Multicultural Community Center, and several identity-specific community resources and groups
- **Academic tutoring:** Student Learning Center: https://slc.berkeley.edu/home
- **Responding to distress or conflict**
  - Resource from AAPI student development: "Maintaining Your Grades in a Mental Health Crisis"
  - Restorative Justice Center: https://rjcenterberkeley.org/